2019 IEEE Intelligent Transportation Systems Conference (ITSC)
Auckland, NZ, October 27-30, 2019

Occupancy Grids Generation Using Deep Radar Network for Autonomous Driving

Nick Engelhardt[1] and Rodrigo Pérez[2] and Qing Rao[1]

Abstract— **Artificial intelligence and deep learning are revolutionizing modern society. While deep-learning-based approaches gradually start dominating the area of the 2D image and 3D point cloud processing, traditional rule-based algorithms remain the mainstream solution in radar signal processing. To unify the approaches for image, radar, and lidar processing for autonomous driving and take advantage of neural networks, the automotive industry is actively seeking solutions for integrating radar into a deep-learning-based sensor fusion framework. In this paper, we propose deep neural networks that take raw radar data as input and generate occupancy grids within the radar's field of view. Furthermore, we integrate the developed system in a test vehicle and demonstrate it on public roads. The proposed networks are the first step towards solving a multiple target detection problem with radar data using deep learning approach.**

I. INTRODUCTION

The automotive industry is developing at full throttle towards highly and fully autonomous driving. One of the essential tasks for an autonomous vehicle is being able to perceive and understand its 360-degree surroundings at a semantic level. To approach this challenging task, different types of sensors such as camera, radar, and lidar are equipped to self-driving prototypes, covering their surroundings without any blind spots. Among them, radar is considered to be the only sensor type that is capable of directly capturing the dynamics of surrounding objects, e.g., the speed and moving direction of other road users. Also, radar sensors are characterized by their high reliability and robustness in harsh environments. It has already been widely used globally in series production cars for Advanced Driver Assistance Systems (ADAS) features such as adaptive cruise control and active braking assist. For near-future autonomous vehicles, radar is an indispensable type of sensor.

In the current generation of series production cars, the core algorithms of the ADAS system are still "rule-based", which are characterized by hand-crafted features, manually adjusted thresholds, and empirically designed logics. The rule-based approaches are proven to be effective and efficient in solving simple problems, such as driving at a constant speed or keeping the ego-car in the lane. However, for highly and fully autonomous driving, especially in urban areas, these are no longer capable of coping with complex problems, e.g., "yield in front of a crosswalk if a pedestrian intends to cross the road". The recent boom of artificial intelligence and deep

[1]BMW Group, Alfred-Nobel-Str. 3, 85716 Unterschleißheim, Germany `firstname.lastname@bmw.de`
[2]Technical University of Munich, Arcisstr. 21, 80333 Munich, Germany `rodrigo.perez@tum.de`

Fig. 1: Our test vehicle with Radarbook, camera, and lidar.

learning in both academia and industry made it possible to tackle this kind of problems using deep neural networks.

So far, environment perception and scene understanding problems have already been extensively researched using cameras, lidar sensors, and combinations of these as inputs to deep neural networks [1], [2], [3]. Using data acquired by radar sensors, however, is considered not well studied yet, partially due to limited availability of public datasets. As a side effect, recent deep-learning-based sensor fusion algorithms for autonomous driving inclined to focus more on camera and lidar rather than radar sensor [4]. How to make use of the information measured by radar for deep neural networks? Which kind of representation of radar measurements are most suitable for deep learning? These are the research questions we address in this work.

In this paper, we present deep neural networks that take raw radar data as input and generate occupancy grids within the radar's field of view. The main contribution of our work is, in a nutshell, the proof of concept of deep-learning-based object detection using radar for autonomous driving. From a technical point of view, our contributions are threefold. First, we propose two different representations of radar targets for deep neural networks, namely *occupancy grid representation* and *frustum representation*, respectively, and we explain our design rationales behind them. Second, we collect raw radar data using a test vehicle, and we prepare a small dataset for training with a self-developed semi-automatic labeling tool. Moreover, we integrate the radar system into a test vehicle and demonstrate the proposed deep radar network on public roads. We consider our work to be the first step towards a scalable deep-learning-based sensor fusion solution for autonomous driving and thus highly relevant for the automotive industry.

The rest of this paper is structured as follows. In Section II, we state the exact problems that we study in this paper, with a brief review of related work. In Section III, we propose deep radar networks with two different approaches to representing radar targets, namely the occupancy grid representation (Section III-A) and the frustum representation (Section III-B), respectively. In Section IV, we present the details of in-vehicle integration of our radar system, including hardware modification (Section IV-A) and sensor calibration (Section IV-B). In Section V, we present the experiment results of our proposed deep radar networks, with an in-depth discussion based on the results. In the last section, Section VI, we summarize this paper and give several directions for future research.

II. Problem Statement

A. Object Detection Using Radar

The radar system chosen for this work is the *Radarbook* from INRAS, which is a radar evaluation platform with a $77\,\mathrm{GHz}$ front-end. The reason for choosing this platform instead of a current series sensor is that it provides access to the raw radar data, i.e., the digital intermediate frequency (IF) signal. Additionally, it enables us to freely choose the system parameters and thus adapt them to our requirements.

A chirp sequence frequency modulation is chosen for the radar waveform. For each measurement frame, a sequence of 256 linear frequency ramps (chirps) is transmitted. Simultaneously, the backscattered signals are sampled 320 times at each of the eight receiver channels. A single measurement frame, therefore, comprises a 3-dimensional array of size $320 \times 256 \times 8$. The resulted IF-signal from a single target can be modeled by Eq. 1.

$$s_{\mathrm{B}}(k,l,u) = A \cdot \exp\left(j2\pi \cdot \left(f_{\mathrm{B}} \cdot \frac{k}{f_{\mathrm{s}}} + f_{\mathrm{D}} \cdot l \cdot T_{\mathrm{c}} - f_{\theta} \cdot u\right)\right) \quad (1)$$

Here, k stands for the sample index within one chirp, l for the chirp index, and u for the receiver index. The beat frequency f_{B}, Doppler frequency f_{D}, and normalized spatial frequency f_{θ} are respectively proportional to the target's range, relative radial velocity, and azimuth angle. A is the signal amplitude, T_{c} the chirp duration, and f_{s} the sampling frequency. A more detailed description of Eq. 1 is given in [5]. The unprocessed IF-signals at every receiver during a single frequency chirp are depicted in Fig. 2.

Classically, the first step to process radar data is to apply Fast Fourier Transforms (FFTs) across each one of its dimensions. In second place comes the detection, e.g., Constant False Alarm Rate (CFAR) [6] detection. Detected points can then be clustered into groups, from which object hypothesis and other high-level functions, such as classification, can be implemented. Each of these steps makes assumptions, e.g., about the probability distribution of noise, or uses hand-crafted models. With every subsequent step, a higher abstraction level is reached, and with it, information might be discarded.

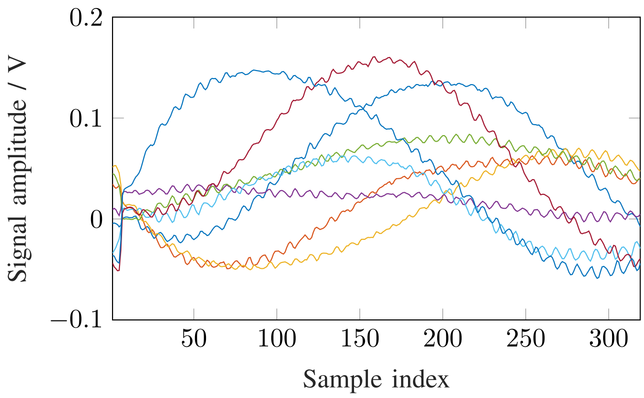


Fig. 2: Signals at the eight receivers during a single frequency chirp.

B. Related Work

Deep learning has been mainly applied in the field of radar signal processing for high-level classification tasks, e.g., in [7] where a system to extract and classify parked vehicles is proposed. Their approach extracts snippets from a radar amplitude grid map and feeds them into a deep convolutional neural network, which then classifies them as either *vehicle* or *non-vehicle*. For road users classification (pedestrians, cars, cyclists, etc.), an approach based on the radar spectrum that uses convolutional neural networks was presented in [5], however, it is limited to moving subjects and is not capable of handling multi-target scenarios. Other works opt for first extracting features from the radar data and then feeding them into a machine learning model to perform the classification [8], [9]. A different approach to classify road users and static objects, which works by performing semantic segmentation on radar point clouds, is presented in [10]. There, the radar point clouds are represented by 4 values (two spatial coordinates, the compensated radial velocity, and the radar cross-section), which are obtained after performing a CFAR detection procedure.

Previous work on radar occupancy grid map representations have often been based on clustered radar detections and sensor models [11], [12], while our work seeks to exploit the power of deep neural networks. To the best of our knowledge, target detection based on raw radar data with deep learning has not been studied so far. There is one work [13], which studied the use of deep learning for automatic target detection on the radar range-Doppler spectrum. Nonetheless, only simulated radar data were considered.

III. Deep Radar Network

The network we propose is shown in Fig. 3. The input layer contains the measurement points of the sampled radar IF-signal, with 81 920 samples per frame. For each of the eight receiving antennas, we extract features from the raw signal through one-dimensional convolutional filters and concatenate them into a feature map. The feature map is connected to the output via two densely connected layers.

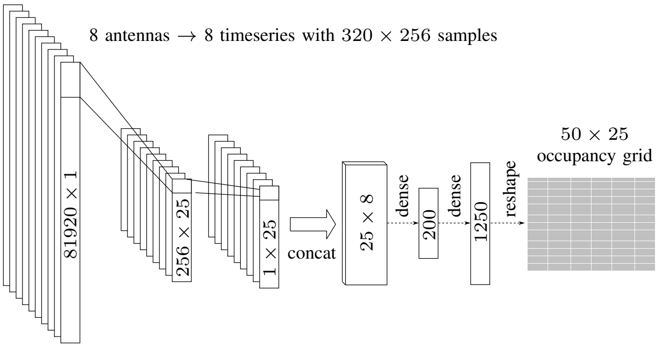
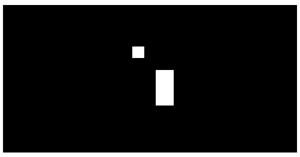


Fig. 3: Network architecture of our proposed deep radar network.

(a) Camera.



(b) Occupancy grid.

(c) Frustum.

Fig. 4: Grid and frustum representation.

A. Occupancy Grid Representation Using 2D Bounding Boxes

Generally, object detection in an image works based on the following two assumptions. i) Every object in the image has a finite extent. ii) The boundaries that distinguish the object from the background or other objects can be identified. In raw radar data, however, these two assumptions do not hold anymore. Unlike an image where semantic information can be retrieved directly within the spatial domain, the information used to identify an object in a radar frame is contained in time-domain signals. More specifically, the range, velocity, and angle of an object are hidden in the IF-signal (Eq. 1). Therefore, we decide not to use bounding box parameterization which is widely used in the context of image object detection for the output of our deep radar network.

Instead of using a fixed number of bounding boxes, we create an occupancy grid that resembles the field of view of the radar device in which objects can be identified. We use a $50m \times 25m$ grid with $1m \times 1m$ cell to represent the sensing area of the Radarbook, which, with our selected parameters, has a maximum sensing range of $25m$. The grid can be represented as a 1250-dimensional vector, where each entry is either *occupied* or *unoccupied*. We consider the task of object detection to be a non-exclusive classification problem of the grid and use the cross-entropy loss in Eq. 2 for training the deep radar network.

$$H(p, q) = -\sum_{i \in N} p_i \cdot \log_2(q_i) \tag{2}$$

In practice, since our training set is heavily imbalanced (much more free cells than occupied cells), we use a weighted cross-entropy loss function. This prevents the network from being trained biased towards predicting fewer occupied cells.

Figure 4b shows an example of the grid representation of the scene in Fig. 4a. Two targets can be identified from the grids. The bigger rectangle represents the red van on the right-hand side, whereas the smaller square the bicyclist on the left-hand side.

Although object velocities can also be retrieved from the range-Doppler spectrum measured by the Radarbook (see Fig. 5), we decide to exclude it at this stage from our deep radar network mainly due to the difficulty getting groundtruth velocity data.

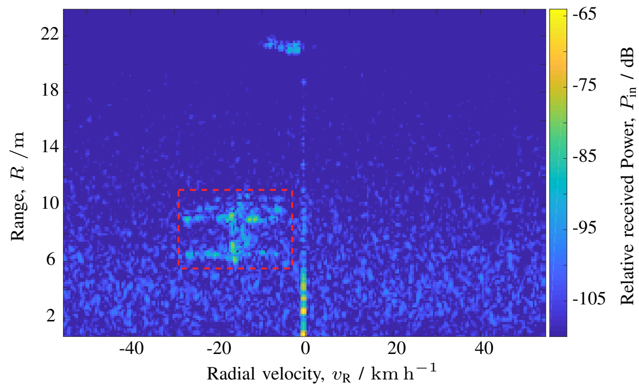


Fig. 5: An example of a range-Doppler power spectrum with a car enclosed in the red rectangle.

B. Frustum Representation

The frustum parameterization captures the range and angle information of objects contained in the radar measurements. Here, we describe each object by its bounding frustum, the portion of a cone or pyramid that lies between two parallel planes cutting it. The near range is calculated using $r_{min} = \sqrt{x_{obj}^2 + y_{obj}^2}$, and the far range $r_{max} = 25m$ is the same as the sensing range of the Radarbook. We calculate the minimum and maximum angle that the frustum occupies from the radar's perspective via the outermost corners of the corresponding object. All cells that lie within these angles and between the planes defined by r_{min} and r_{max} are marked as occupied in the occupancy grid. Figure 4c shows an example of the frustum representation of the same scene as in Fig. 4b. Note that the number of occupied cells is much higher than that using the grid representation. This yields a better balance between occupied and non-occupied cells in the training data, i.e., a less biased network after training.

IV. IN-VEHICLE SYSTEM INTEGRATION

We integrate our radar system into a BMW 3 Series test vehicle. The hardware components relevant to autonomous driving are shown in Fig. 1. We decide to mount the Radarbook in the front middle of the test vehicle to utilize its field of view to the maximum extent possible.

A. Hardware Modification

To mount the Radarbook to an ideal position in the front middle of the test vehicle, both the case of the Radarbook and the BMW "Kidney" grille need to be modified. We design a new case with an extra inlet for the Ethernet connection and power supply (Fig. 6a) which exactly fits into the 3 Series grille (Fig. 6b). Furthermore, we add an opening for a small electric fan to keep the Radarbook in safe operating temperature.

(a)

(b)

Fig. 6: (a) Radarbook with the self-designed case. (b) mounted into the BMW Kidney grille.

B. Calibration

The pose (position and orientation) of the Radarbook with respect to the vehicle coordinate system has to be determined so that its measurements can be used for autonomous driving functions. To achieve this, the translation and rotation between the Radarbook and the middle of the vehicle's rear-axis need to be determined. Here, instead of directly calibrating the Radarbook to the vehicle rear-axis, we perform a cross-sensory calibration between the Radarbook and lidar since the lidar sensors are already calibrated to the vehicle coordinate system.

For the cross-sensory calibration, we use a calibration target (Fig. 7) which can be identified by both the Radarbook and lidar. The calibration target is placed at approximately the same height as the mounting point of the Radarbook, which allows us to skip the calculation of the elevation angle of the target during calibration. Thus, the calibration parameters to be determined comprise a two-dimensional translational offset $(x_{\text{trans}}, y_{\text{trans}})$ and a single rotation angle β around the vertical axis. With the corresponding coordinates from two measurements of the target in both the Radarbook and the vehicle coordinate system, the rotation angle β can be determined through Eq. 3, where a, b, c, and d are placeholder variables whose expressions are given in Eq. 4. r and θ indicate the range and the angle of the target in the Radarbook coordinate system, respectively (Fig. 8).

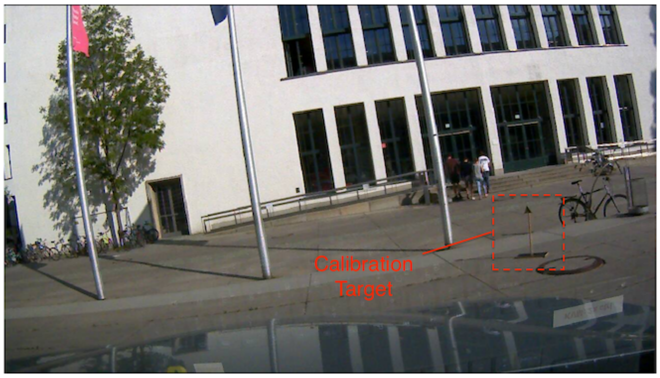


Fig. 7: Corner reflector as a calibration target.

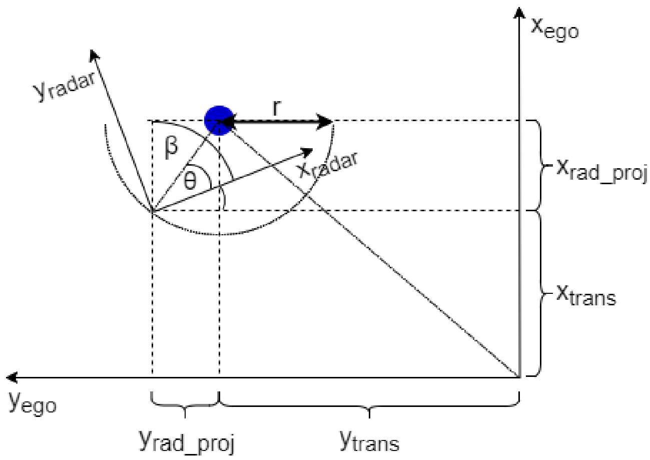


Fig. 8: Geometric transformation from the Radarbook coordinate system to the vehicle rear-axis.

$$\beta = \arcsin\left[\frac{x_{\text{ego}} - x'_{\text{ego}}}{c-a} - \frac{d}{b}\cdot\frac{y_{\text{ego}} - y'_{\text{ego}}}{c-a}\right] \tag{3}$$

$$\begin{aligned} a &= -r\cos(\theta) + r'\cos(\theta') \\ b &= r\sin(\theta) + r'\sin(\theta') \\ c &= r\sin(\theta) - r'\sin(\theta') \\ d &= r\cos(\theta) - r'\cos(\theta') \end{aligned} \tag{4}$$

Subsequently, the translational offset can be calculated using Eq. 5.

$$\begin{aligned} x_{\text{trans}} &= x_{\text{ego}} - r\sin\left(\frac{\pi}{2} - \beta + \theta\right) \\ y_{\text{trans}} &= y_{\text{ego}} + r\sin(\beta - \theta) \end{aligned} \tag{5}$$

V. EXPERIMENTAL RESULTS

A. Dataset

We use a self-generated dataset to evaluate our proposed deep radar networks. The dataset comprises approximately 500 frames of raw measurements from the Radarbook, mainly recorded in urban scenarios with heavy traffic. We split the dataset into three subsets for training, validation, and testing (evaluation), respectively. Also, we record lidar point clouds and camera images (RGB) for the groundtruth generation.

For the object detection task for autonomous driving, a groundtruth object label should at least contain a two-dimensional bounding box in the bird's-eye view and an object class label. We developed a labeling tool with a graphical user interface for manually generating labels. Radar, lidar and camera measurements are used for manual annotation, as shown in Fig. 9. Furthermore, we cluster the radar measurements using DBSCAN [14] which generates candidate groundtruth bounding boxes before the manual labeling process. This increases the level of automation and speeds up the entire groundtruth generation process.

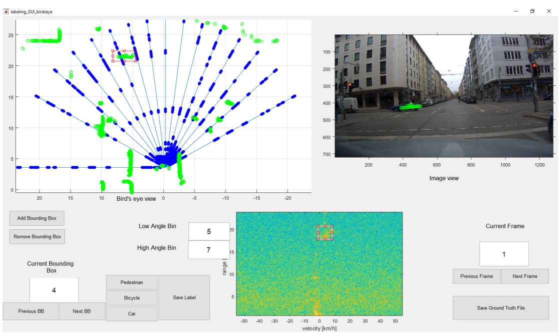


Fig. 9: The graphical user interface of our self-developed labeling tool. The green points are lidar measurements which help a human labeler identifying groundtruth objects.

B. Network Evaluation

We evaluate three variants of proposed deep radar networks using precision-recall, which is one of the standard evaluation metrics for object detection. The results are presented in Tab. I and Fig. 10. *Grid* refers to the proposed network using occupancy grid representation (Sect. III-A), while *Frustum* indicating the frustum representation (Sect. III-B). *FC* and *Conv* describe the last layers of the network architecture, namely fully-connected and convolutional, respectively. We use the conventional radar data processing pipeline (Sect. II-A) as the baseline algorithm for the comparison. The red '+' in Fig. 10 marks the result of the baseline algorithm, which shows approximately 6% precision by around 50% recall. The evaluation results clearly show that all three developed variants outperform the baseline algorithm. Among them, the Frustum+FC variant yields the best precision-recall performance.

Method	Average		Maximum	
	Precision	Recall	Precision	at Recall
Baseline	-	-	6.18%	49.35%
Grid+FC	19.79%	32.30%	20.55%	28.67%
Grid+Conv	14.30%	10.34%	14.84%	9.48%
Frustum+FC	31.24%	39.13%	32.31%	30.80%

TABLE I: Precision-recall performance of our proposed deep radar networks. The *Maximum* column shows when the network reaches its maximum precision performance.

We believe that the frustum variant outperforms the others mainly because it resembles the natural radial-angular structure of radar measurements. Besides, in a radar frame, the rear edges of an object are mostly occluded, i.e., it is impossible to estimate the complete outline of the object only using radar measurement. Forcing a neural network to learn rectangular bounding boxes from a radar frame is an over-constrained problem. We believe that this is the main reason why the maximum recalls of the two variants using grid representation only reached around 40%.

Still, the performance of all our proposed deep radar networks needs to be boosted. We expect improved performance once we have sufficient groundtruth data for training.

C. Road Demonstration

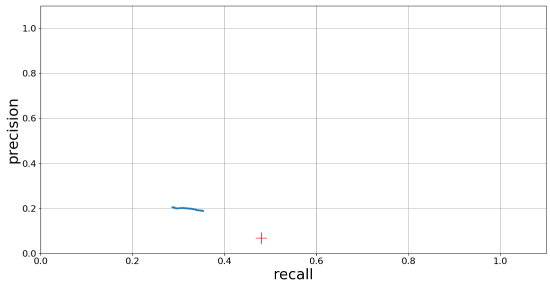
We demonstrate our proposed deep radar networks in urban traffic. Figure 11 shows the rviz [1] user interface displayed in the co-driver monitor installed in our test vehicle. The output of the network (using the frustum representation) is shown on the left, and the camera image is displayed on the right. White cells in the output indicate occupied area whereas black cells free space. In the scene shown in Fig. 11, the network detects the car in front of the ego-car and assumes the cone area behind it is non-drivable.

During the road demo, we observed that the output of the network was reasonable in situations that a human driver would consider "simple". However, if the traffic scenario became complicated, the output started to diverge from reality. We believe this phenomenon is mainly due to the limited training dataset which does not contain many complicated traffic scenarios. Still, we consider the concept of deep radar network to be proved since training with less than 500 frames already yielded reasonable outputs during road demonstration.

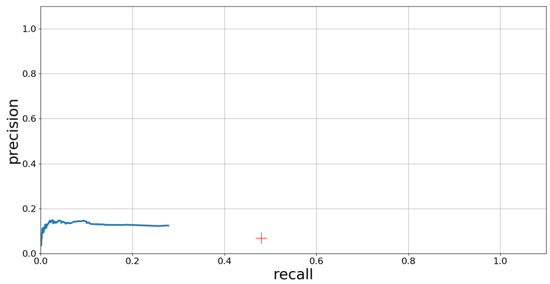
VI. CONCLUDING REMARKS

In this paper, we presented deep radar networks as the first step towards a unified deep-learning-based sensor fusion solution for autonomous driving. We proposed two representations of raw radar measurements for deep neural networks, namely *occupancy grid representation* and *frustum representation*, and we discussed our design rationales behind them. By integrating the radar system into a test vehicle and demonstrating it on public roads, we proved the concept of object detection in the surroundings of the ego-car using our proposed deep radar networks. The experimental

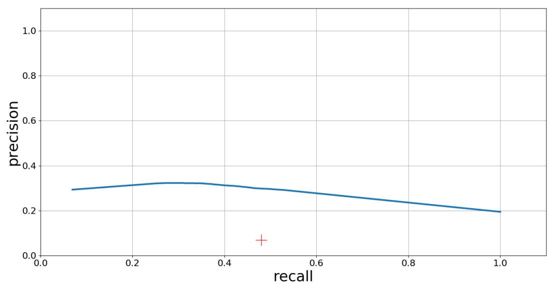
[1] Robot Operating System (ROS) Visualization.



(a) Grid+FC.



(b) Grid+Conv.



(c) Frustum+FC.

Fig. 10: Evaluation of proposed deep radar networks.

results showed better performance compared to the baseline algorithm, which still needs to be boosted due to the limited size of the dataset used for training. We believe that with a larger amount of training data, the performance of our proposed deep radar networks will be significantly improved.

Apart from acquiring more data, our future directions of research include further experimenting with different network architectures of deep radar networks and enabling multi-object tracking. These are essential for a unified deep-learning-based sensor fusion solution. We will devote ourselves to improving the performance of the deep radar network and making it practical for the implementation in series production cars.

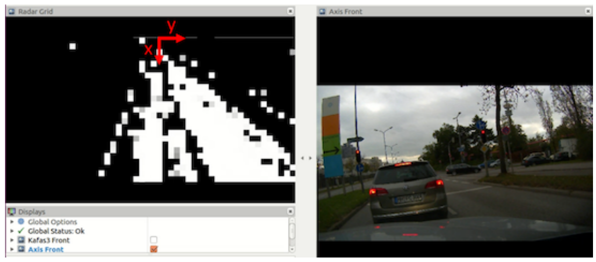


Fig. 11: Demonstration of deep radar networks in urban traffic. The x-axis in the output points to the driving direction, while the y-axis points to the left from the driver's perspective.

References

[1] J. Ku, M. Mozifian, J. Lee, A. Harakeh, and S. Waslander. (2017) Joint 3D Proposal Generation and Object Detection from View Aggregation. arXiv. [Online]. Available: https://arxiv.org/abs/1712.02294

[2] J. Kim, J. Koh, Y. Kim, J. Choi, Y. Hwang, and J. W. Choi, "Robust Deep Multi-modal Learning Based on Gated Information Fusion Network," in *Asian Conference on Computer Vision (ACCV)*, 2018, p. to appear.

[3] M. Liang, B. Yang, S. Wang, and R. Urtasun, "Deep Continuous Fusion for Multi-Sensor 3D Object Detection," in *European Conference on Computer Vision (ECCV)*, 2018, pp. 663–678.

[4] S. Agarwal, J. Ogier du Terrail, and F. Jurie. (2018) Recent Advances in Object Detection in the Age of Deep Convolutional Neural Networks. arXiv. [Online]. Available: https://arxiv.org/abs/1809.03193

[5] R. Pérez, F. Schubert, R. Rasshofer, and E. Biebl, "Single-Frame Vulnerable Road Users Classification with a 77 GHz FMCW Radar Sensor and a Convolutional Neural Network," in *International Radar Symposium (IRS)*, 2018, pp. 1–10.

[6] H. Rohling, "Radar CFAR Thresholding in Clutter and Multiple Target Situations," *IEEE Transactions on Aerospace and Electronic Systems*, vol. AES-19, no. 4, pp. 608–621, 1983.

[7] J. Lombacher, M. Hahn, J. Dickmann, and C. Wöhler, "Detection of Arbitrarily Rotated Parked Cars Based on Radar Sensors," in *International Radar Symposium (IRS)*, 2015, pp. 180–185.

[8] R. Prophet, M. Hoffmann, M. Vossiek, C. Sturm, A. Ossowska, W. Malik, and U. Lübbert, "Pedestrian Classification with a 79 GHz Automotive Radar Sensor," in *International Radar Symposium (IRS)*, 2018, pp. 1–6.

[9] S. Heuel and H. Rohling, "Pedestrian Classification in Automotive Radar Systems," in *International Radar Symposium (IRS)*, 2012, pp. 39–44.

[10] O. Schumann, M. Hahn, J. Dickmann, and C. Wöhler, "Semantic segmentation on radar point clouds," in *2018 21st International Conference on Information Fusion (FUSION)*, July 2018, pp. 2179–2186.

[11] A. Loeffler, J. Ronczka, and T. Fechner, "Parking Lot Measurement with 24 GHz Short Range Automotive Radar," in *International Radar Symposium (IRS)*, 2015, pp. 137–142.

[12] K. Werber, M. Rapp, J. Klappstein, M. Hahn, J. Dickmann, K. Dietmayer, and C. Waldschmidt, "Automotive Radar Gridmap Representations," in *IEEE MTT-S International Conference on Microwaves for Intelligent Mobility (ICMIM)*, 2015, pp. 1–4.

[13] L. Wang, J. Tang, and Q. Liao, "A study on radar target detection based on deep neural networks," *IEEE Sensors Letters*, vol. 3, no. 3, pp. 1–4, March 2019.

[14] M. Ester, H. P. Kriegel, J. Sander, and X. Xu, "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise," in *International Conference on Knowledge Discovery and Data Mining (KDD)*, 1996, pp. 226–231.